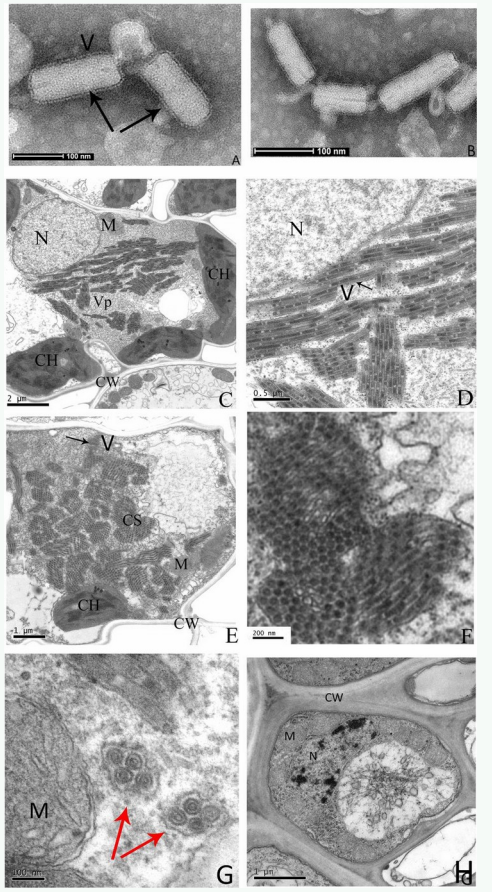
Introduction to Photosynthesis

- Photosynthesis is the primary process by which autotrophic organisms convert light energy into chemical energy [S1].
- This biological process occurs within specialized organelles known as chloroplasts found in plant cells [S1].
- The overall reaction involves taking in carbon dioxide and water to produce glucose and oxygen.
- It serves as the foundational energy source for nearly all life forms on Earth.
- The process is divided into two main stages: light-dependent reactions and the Calvin cycle [S1].



Fmicb-07-02140-g002 — Xin Yang, Jilei Huang, Chuanhe Liu, Biao Chen, Tong Zhang, and Guohui Zhou (CC BY 4.0), Wikimedia Commons

Agenda

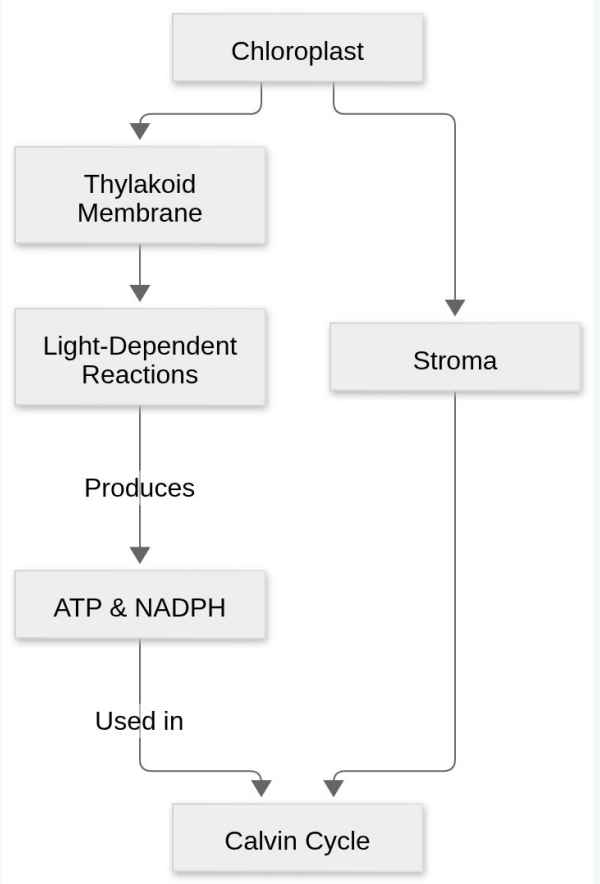
- Overview of chloroplast structure and its role in energy capture.
- Detailed mechanics of the light-dependent reactions in the thylakoid.
- Analysis of the Calvin cycle and carbon fixation in the stroma.
- The role of key molecules like ATP, NADPH, and the enzyme RuBisCO [S1].
- A worked example of the photosynthetic chemical equation.
- Summary of energy flow and a final knowledge check.

Learning Objectives

- Identify the specific locations of the two main stages of photosynthesis within the chloroplast [S1].
- Describe the process of water splitting and oxygen release during light reactions [S1].
- Explain the function of ATP and NADPH as energy carriers between reactions [S1].
- Understand how the enzyme RuBisCO facilitates carbon dioxide fixation [S1].
- Calculate the inputs and outputs required for the synthesis of one glucose molecule.

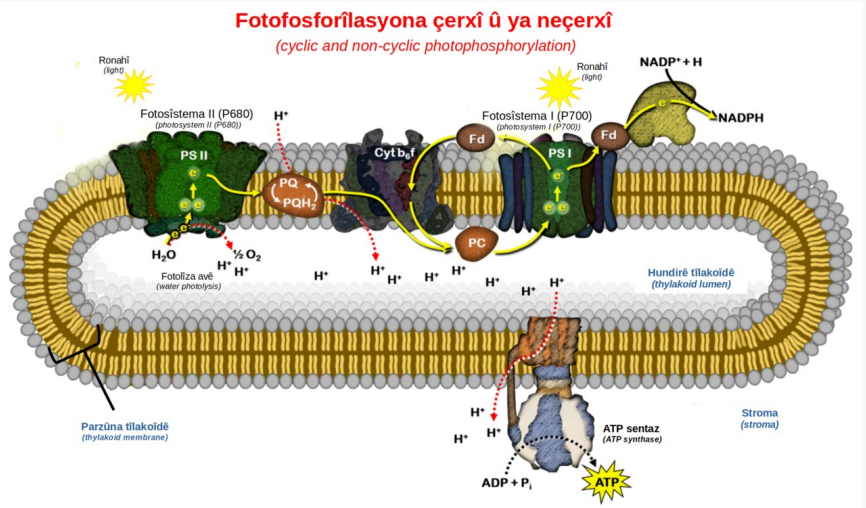
The Chloroplast Structure

- Chloroplasts are double-membrane organelles containing a fluid called stroma [S1].
- Thylakoids are flattened sac-like structures arranged in stacks called grana.
- The thylakoid membranes contain chlorophyll pigments that absorb light energy.
- Light-dependent reactions are physically localized within these thylakoid membranes [S1].
- The stroma provides the site for the light-independent reactions, or the Calvin cycle [S1].



Light-Dependent Reactions

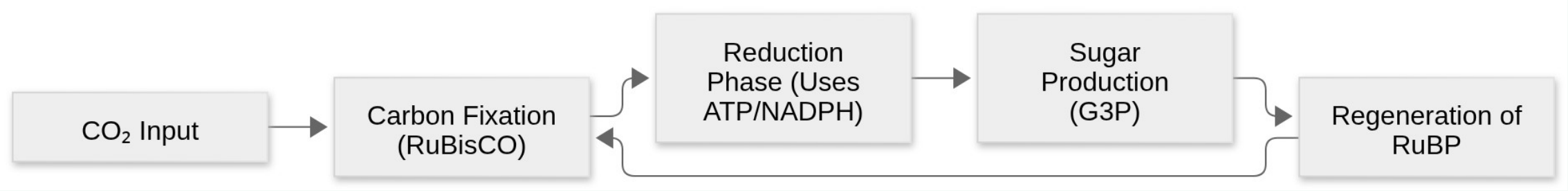
- These reactions require direct sunlight to excite electrons in chlorophyll.
- Water molecules (H_2O) are split in a process called photolysis to replace lost electrons [S1].
- The splitting of water results in the release of oxygen (O_2) as a byproduct [S1].
- Solar energy is converted into chemical energy in the form of ATP and NADPH [S1].
- These energy-rich molecules are then transported to the stroma for the next phase.



Light reaction cyclic & linear flow ku — ELaurent (CC BY-SA 4.0), Wikimedia Commons

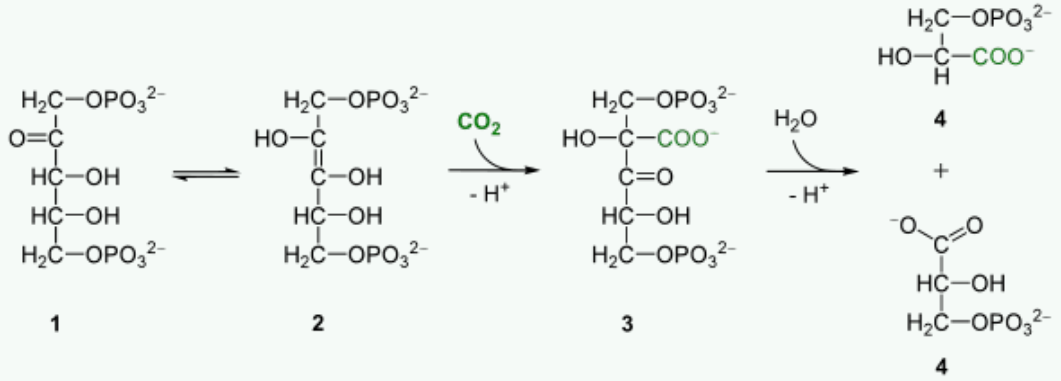
The Calvin Cycle (Light-Independent Reactions)

- The Calvin cycle takes place in the stroma and does not directly require light [S1].
- It utilizes the ATP and NADPH produced during the light-dependent reactions [S1].
- Carbon dioxide (CO_2) from the atmosphere is "fixed" into organic molecules [S1].
- The process involves a series of enzyme-mediated steps to eventually produce glucose.
- This cycle must turn multiple times to produce a single stable sugar molecule.



The Role of RuBisCO

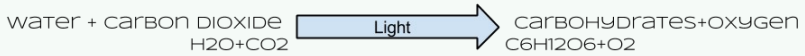
- RuBisCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase) is the enzyme that catalyzes the first step of the Calvin cycle [S1].
- It attaches CO_2 to a five-carbon sugar called RuBP [S1].
- This step is known as carbon fixation because it "fixes" gas into a solid form.
- RuBisCO is relatively slow and inefficient, so plants produce it in massive quantities.
- Without this specific enzyme, life as we know it would not have a way to bring carbon into the food chain.



Rubisco com carboxilasa — Yikrazuul (CC BY-SA 4.0), Wikimedia Commons

Worked Example: The Photosynthetic Equation

- The balanced chemical equation is: $6CO_2 + 6H_2O + \text{Light Energy} \rightarrow C_6H_{12}O_6 + 6O_2$.
- To produce 1 molecule of glucose ($C_6H_{12}O_6$), the plant requires 6 molecules of carbon dioxide.
- 6 molecules of water are split to provide the necessary electrons and protons [S1].
- This process results in 6 molecules of oxygen being released into the atmosphere [S1].
- The energy stored in the bonds of glucose comes originally from the captured photons.



Photosynthesis equation with words and symbols — Fortninja (CC0), Wikimedia Commons

Recap of Key Concepts

- Photosynthesis occurs in the chloroplast: light reactions in thylakoids, Calvin cycle in stroma [S1].
- Light reactions split water and release oxygen while creating ATP and NADPH [S1].
- The Calvin cycle uses that ATP and NADPH to fix CO_2 into glucose [S1].
- RuBisCO is the essential enzyme for capturing atmospheric carbon [S1].
- The process converts solar energy into stable chemical energy stored in sugars.

Discussion and Assessment

- How would a lack of water affect the production of ATP and NADPH?
- Why is the Calvin cycle often called the "Dark Reactions" even though it usually happens during the day?
- Predict what would happen to a plant if the enzyme RuBisCO was inhibited by a toxin.
- Explain the relationship between the oxygen we breathe and the process of photolysis.
- Discuss how light intensity might change the rate of glucose production.



Green leaf with sunlight — Joselodos (CC0), Wikimedia Commons